

SQUIRE

Intelligent Task Management System Using Natural Language
Understanding (NLU)

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1 Abstract

This project presents an intelligent task management system that enables users to manage tasks, meetings, and notes using natural language queries. The system leverages Natural Language Understanding (NLU) with transformer-based models to interpret user intent and extract key entities such as dates, times, and task details. A multi-task classification approach is employed to identify both the action (Add, Get, Update, Delete) and the object (Task, Meeting, Note, Progress). The extracted information is processed through a backend system that converts user input into structured database operations and triggers automated scheduling in calendar services. The system aims to enhance productivity by providing an efficient and user-friendly conversational interface.

2 Project Goals

- Develop an intelligent task management system supporting natural language interaction.
- Build an NLU model for action and object classification.
- Design an entity extraction module for dates, times, and task details.
- Implement a backend system to convert natural language into database operations.
- Integrate automated scheduling with calendar services.
- Develop a mobile application for user interaction.
- Ensure scalability and extensibility for future improvements.

3 Methodology and Technologies

3.1 Natural Language Understanding (NLU)

The system utilizes transformer-based models to process user input. A multi-task learning approach is adopted to classify both actions (Add, Get, Update, Delete) and objects (Task, Meeting, Note, Progress). Additionally, an entity extraction component identifies relevant information such as task titles, dates, and participants.

3.2 Backend Development

A backend API is developed to handle user requests and process the outputs of the NLU model. A dynamic routing mechanism maps actions and objects to appropriate database operations efficiently.

3.3 Automation and Scheduling

The system supports automated scheduling by detecting time-related entities and creating corresponding events in external calendar services.

3.4 Deployment & MLOps

The system is deployed as a mobile-based application supported by a scalable backend. Model versioning, monitoring, and API management are implemented to ensure reliability and maintainability.

4 System Workflow

1. User submits a natural language query.
2. The NLU model extracts action, object, and entities.
3. Date and time expressions are converted into structured formats.

4. The backend maps the output to database operations.
5. Data is stored or retrieved accordingly.
6. Automation triggers scheduling when required.
7. The system returns a response to the user.

5 Project Planning & Management

5.1 Project Proposal

The project aims to develop a smart task management system that simplifies user interaction through natural language and automates scheduling processes.

5.2 Project Plan

- Dataset preparation and preprocessing
- Model training and evaluation
- Backend API development
- Development of mobile applications
- Integration and testing of systems

5.3 Task Assignment & Roles

- NLP Development: Intent classification and entity extraction
- Backend Development: API design and database management
- Mobile Development: User interface and system integration
- N8N: Connect to Google calendar
- RAG: for unstructured data like notes

5.4 Risk Assessment & Mitigation

- Limited dataset → improve data collection and preprocessing
- Model performance issues → fine-tuning and optimization
- Integration challenges → modular architecture and testing
- Time constraints → clear milestones and task distribution

5.5 Key Performance Indicators (KPIs)

- Intent classification accuracy
- Entity extraction performance (F1-score)
- API response time
- System uptime and reliability
- Task scheduling success rate

6 Deliverables

- Mobile application
- Backend API
- Trained NLU model
- RAG
- Database system
- Calendar integration
- Final documentation and report

7 Impact

The proposed system improves productivity by enabling users to manage tasks efficiently through natural language. It reduces manual effort, enhances usability, and demonstrates the practical application of NLU in real-world systems.